

PPL Week-7

Rachit Takate – 230962294 – B54

I. Write a program in CUDA to count the number of times a given word is repeated in a sentence. (Use atomic function)

```
#include <cuda.h>
#include <stdio.h>
#include <string.h>

__global__ void countSubStr(char *dSent, char *dWord, int *dWordcount, int sLen, int wLen) {
    int i = blockIdx.x * blockDim.x + threadIdx.x;

    if (i + wLen <= sLen) {
        int j;

        for(j = 0; dSent[i + j] == dWord[j] && j < wLen; j++);

        if(j == wLen && (i + j == sLen || dSent[i + j] == ' '))
            atomicAdd(dWordcount, 1);
    }
}

int main() {
    int wCount = 0;
    char sent[64], word[16];
    char *dSent, *dWord;

    printf("Enter sentence: ");
    scanf("%[^\n]*c", sent);
    printf("Enter word:");
    scanf("%[^\n]*c", word);

    int *dWordcount, sLen = strlen(sent),
        wLen = strlen(word);
    size_t sSize = sizeof(char) * sLen,
        wSize = sizeof(char) * wLen;

    cudaMalloc((void **) &dSent, sSize);
    cudaMalloc((void **) &dWord, wSize);
    cudaMalloc((void **) &dWordcount, sizeof(int));

    cudaMemcpy(dSent, sent, sSize, cudaMemcpyHostToDevice);
    cudaMemcpy(dWord, word, wSize, cudaMemcpyHostToDevice);
    cudaMemcpy(dWordcount, &Wordcount, sizeof(int), cudaMemcpyHostToDevice);

    countSubStr<<<1, sSize>>>(dSent, dWord, dWordcount, sLen, wLen);

    cudaMemcpy(&Wordcount, dWordcount, sizeof(int), cudaMemcpyDeviceToHost);
    printf("Count: %d\n", wCount);

    cudaFree(dWordcount);
    cudaFree(dWord);
    cudaFree(dSent);

    printf("Rachit 54\n");
}
```

```

return 0;
}

```

Output

```

C:\D_Drive\MAHE\6thSem\PPL> nvcc Q1.cu
C:\D_Drive\MAHE\6thSem\PPL> ./a.out
Enter sentence: hi hello hey hi world hi
Enter word: hi
Count: 3
Rachit 54

```

II. Write a CUDA program that reads a string S and produces the string RS as follows:

Input string S: PCAP Output string RS: PCAPPCAPCP

Note: Each work item copies required number of characters from S in RS.

```

#include <cuda.h>
#include <stdio.h>
#include <string.h>

__global__ void RSGen(char *dS, char *dRS, int Slen, int RSlen){
    int gtid = blockIdx.x * blockDim.x + threadIdx.x;
    if(gtid >= Slen) return;

    int start = 0, temp = Slen;
    for (int i = gtid; i > 0; i--){
        start += temp;
        temp--;
    }
    for (int i = 0; i < Slen-gtid; i++) dRS[start + i] = dS[i];
}

int main() {

    char S[15];
    printf("Enter String S:"); scanf("%s", S);
    int Slen = strlen(S);
    int RSlen = ((Slen*(Slen+1))/2);
    char RS[RSlen+1];

    char *dS, *dRS;
    cudaMalloc((void **)&dS, sizeof(char) * Slen);
    cudaMalloc((void **)&dRS, sizeof(char) * RSlen);

    cudaMemcpy(dS, &S[0], sizeof(char) * Slen, cudaMemcpyHostToDevice);

    RSGen<<<1, Slen>>>>(dS, dRS, Slen, RSlen);

    cudaMemcpy(RS, dRS, sizeof(char) * RSlen, cudaMemcpyDeviceToHost);
    RS[RSlen] = '\0';

    printf("RS String is %s \n", RS);

    cudaFree(dS);
    cudaFree(dRS);

    return 0;
}

```

Output

```
C:\D_Drive\MAHE\6thSem\PPL> nvcc Q2.cu
C:\D_Drive\MAHE\6thSem\PPL> ./a.out
Enter string: PCAP
PCAPPCAPCP
```

III. Write a CUDA program which reads a string consisting of N words and reverse each word of it in parallel.

```
#include <cuda.h>
#include <stdio.h>
#include <string.h>

__global__ void reverseWord(char *dS, int start, int end){
    int gtid = threadIdx.x;
    int wordLen = end - start + 1;
    int mid = wordLen / 2;

    if(gtid < mid) {
        char temp = dS[start + gtid];
        dS[start + gtid] = dS[start + (wordLen - 1 - gtid)];
        dS[start + (wordLen - 1 - gtid)] = temp;
    }
}

int main() {
    char S[100];
    printf("Enter string: ");
    scanf("%s", S);

    int Slen = strlen(S);
    char *dS;
    cudaMalloc((void **)&dS, Slen * sizeof(char));
    cudaMemcpy(dS, S, Slen * sizeof(char), cudaMemcpyHostToDevice);

    for (int i = 0; i < Slen; i++) {
        int start = i;
        while (i < Slen && S[i] != ' ') i++;
        int end = i - 1;

        if (end >= start) {
            int threads = (end - start + 1) / 2;
            if(threads > 0) reverseWord<<<1, threads>>>(dS, start, end);
        }
    }

    cudaMemcpy(S, dS, Slen * sizeof(char), cudaMemcpyDeviceToHost);
    S[Slen] = '\0';
    printf("%s\n", S);

    cudaFree(dS);

    return 0;
}
```

Output

```
C:\D_Drive\MAHE\6thSem\PPL> nvcc A1.cu
C:\D_Drive\MAHE\6thSem\PPL> ./a.out
Enter string: ADCB Hey DOG
BCDA yeH GOD
```

IV. Write a CUDA program that takes a string Str as input and one integer value N and produces an output string , Sout, in parallel by concatenating input string Sin, N times as shown below.

Input: Sin =“Hello” N=3 Ouput: Sout= “HelloHelloHello”

Note: Every thread copies the same character from the Input string S, N times to the required position.

```
#include <stdio.h>
#include <string.h>
#include <cuda.h>

__global__ void concatenateString(char *d_in, char *d_out, int len, int n) {
    int tid = blockIdx.x * blockDim.x + threadIdx.x;

    if (tid < len * n) {
        int sourceIndex = tid % len;
        d_out[tid] = d_in[sourceIndex];
    }
}

int main() {
    char Sin[100];
    int N;

    printf("Enter string: ");
    scanf("%s", Sin);
    printf("Enter N: ");
    scanf("%d", &N);

    int len = strlen(Sin);
    int outLen = len * N;

    char *d_in, *d_out;
    char *Sout = (char *) malloc((outLen + 1) * sizeof(char));

    cudaMalloc((void **) &d_in, len * sizeof(char));
    cudaMalloc((void **) &d_out, outLen * sizeof(char));

    cudaMemcpy(d_in, Sin, len * sizeof(char), cudaMemcpyHostToDevice);

    int threadsPerBlock = 256;
    int blocksPerGrid = (outLen + threadsPerBlock - 1) / threadsPerBlock;

    concatenateString<<<blocksPerGrid, threadsPerBlock>>>(d_in, d_out, len, N);

    cudaMemcpy(Sout, d_out, outLen * sizeof(char), cudaMemcpyDeviceToHost);
    Sout[outLen] = '\0';

    printf("%s\n", Sout);

    cudaFree(d_in);
    cudaFree(d_out);
}
```

```
    return 0;
}
```

Output

[illegible]